Skill Library

Videos

Gemini

Skill card

Goal Fruit on the cake
Plan Open fridge, pick fruit, place
Check Fruit rests on the cake
Keyframes

Input

Task Instruction

Open the cabinet. ..... carry it to the dining counter.

Observation

Visual Memory

History analysis & Select frames

VLM

Episode Visual History

T_0 ... T

Recall

Agentic Planner

Recall memory when needed

"Where was the basket?"

Execution History

Tool Calls, Subtask Results, and Planner Actions

Actor

Subtask VLA

Geometric Skills

.....

Geometric Skills

Accessibility Check



SAM & Depth Information

Reachable → Manipulate

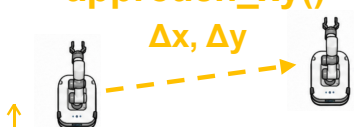
Near limit → approach_xy()

Too far → goto_stance()

Base Adjustment

approach_xy()

Δx, Δy



Local XY shift
Keep heading fixed

goto_stance()



New stance
Avoid obstacles

End-Effector Adjustment



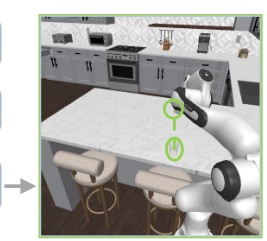
Before adjustment

SAM & Depth Information

3D Object Position

EEF motion | Base motion

Adjust robot pose



After adjustment

Robot Execution Example

ROBOCASA365

Subtask VLA → Accessibility Check → Subtask VLA → Subtask VLA → Base Adjustment → Subtask VLA

